

LESSON 5-4 • SESSION 1 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Area of Triangles

Builds on Lesson 5-2 Session 1 • 6.G.A.1

TODAY'S LESSONYou will find one center triangle, like $\frac{1}{2} \times 16 \times 11$ for the pentagon sign.**YOU NEED TO**Find a triangle's area with $A = \frac{1}{2} \times b \times h$, like $\frac{1}{2} \times 14 \times 9$. **You will use it in Practice Form A, problems 1, 3, and 5.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Write $A = \frac{1}{2} \times b \times h$ with the numbers.
- 2** Halve the even number first.
- 3** Multiply what is left. Write square units.

WATCH ME*the steps in action***A triangle has a base of 14 cm and a height of 9 cm. Find its area.****Step 1** $A = \frac{1}{2} \times 14 \times 9$.**Step 2** Half of 14 is 7.**Step 3** $7 \times 9 = 63$, so 63 cm².**Answer** 63 cm²**TRY IT WITH HELP***fill in each step*

1. Base 10 cm, height 7 cm. Find the area.

Step 1 $A = \frac{1}{2} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$ **Step 2** Half of 10 is $\underline{\hspace{2cm}}$ **Step 3** $5 \times 7 = \underline{\hspace{2cm}}$ cm²

2. Base 8 m, height 6.9 m. Find the area.

Step 1 $A = \frac{1}{2} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$ **Step 2** Half of 8 is $\underline{\hspace{2cm}}$ **Step 3** $4 \times 6.9 = \underline{\hspace{2cm}}$ m²**YOUR TURN***use the steps on your own*

3. Find each area. Use $A = \frac{1}{2} \times b \times h$.

a. $\frac{1}{2} \times 6 \times 5 = \underline{\hspace{2cm}}$

b. $\frac{1}{2} \times 12 \times 6 = \underline{\hspace{2cm}}$

c. $\frac{1}{2} \times 20 \times 9 = \underline{\hspace{2cm}}$

d. $\frac{1}{2} \times 8 \times 11 = \underline{\hspace{2cm}}$

e. $\frac{1}{2} \times 4 \times 3.5 = \underline{\hspace{2cm}}$

f. $\frac{1}{2} \times 10 \times 6.9 = \underline{\hspace{2cm}}$

Multiply Area by Equal Parts

Builds on Grade 4 • 4.NBT.B.5

TODAY'S LESSONYou will find $6 \times 45 \text{ cm}^2$ for Jamal's hexagon tile.**YOU NEED TO**Multiply one part's area by the number of equal parts, like 8×45 . **You will use it in Practice Form A, problems 2, 7, and 8.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Find** the area of one part.
- 2 Count** the equal parts.
- 3 Multiply:** one part $\times$ number of parts.

WATCH ME*the steps in action***A quilt is made of 8 equal triangles. Each is 45 in^2 . Find the quilt's area.****Step 1** One part: 45 in^2 .**Step 2** Number of parts: 8.**Step 3** $8 \times 45 = 360$.**Answer** **360 in^2** **TRY IT WITH HELP***fill in each step***4.** A hexagon is made of 6 equal triangles. Each is 15 cm^2 . Find the total area.**Step 1** One part: _____ cm^2 **Step 2** Number of parts: _____**Step 3** $6 \times 15 =$ _____ cm^2 **5.** A pentagon is made of 5 equal triangles. Each is 24 m^2 . Find the total area.**Step 1** One part: _____ m^2 **Step 2** Number of parts: _____**Step 3** $5 \times 24 =$ _____ m^2 **YOUR TURN***use the steps on your own***6.** Find the total area.**a.** 6 parts of 12 cm^2 _____**b.** 5 parts of 18 m^2 _____**c.** 8 parts of 15 in^2 _____**d.** 6 parts of 35 ft^2 _____**e.** 10 parts of 16 cm^2 _____**f.** 8 parts of 25 m^2 _____**READY CHECK***a problem like the one in today's lesson***7.** A hexagon has 6 center triangles of 40 cm^2 each. What is its area?**A** 40 cm^2 **B** 46 cm^2 **C** 240 cm^2 **D** 480 cm^2 **CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____